

Failure localization with active and passive voltage contrast in FIB and SEM

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Abstract The common Passive Voltage Contrast localization method in Focused Ion Beams and Scanning Electron Microscopes can be used for failure localization issues. These methods became widely accepted in the semiconductor failure analysis community. Nearly all labs make use of it. The Active Voltage Contrast method works with additional external voltages applied inside the chamber to certain structures at the sample surface and offers even more localization possibilities. A comprehensive overview over all phenomena related to Voltage Contrast generation is given and the multiple advantages, possibilities and limits of VC failure localization are systemized and discussed.

1 Introduction

Passive Voltage Contrast (PVC) localization [1–7] is based on brightness differences in FIB or SEM images of more or less conducting structures and can be used for failure localization on semiconductor circuits. The Active Voltage Contrast method (AVC) offers even more localization possibilities. Here an external bias is applied to certain structures so that the contrast differences are more pronounced. The main advantage of the AVC is the fact, that not just opens and shorts can be detected but also failures caused by an increased resistance of conductors or contacts.

2 Passive voltage contrast in FIB

2.1 Basic principles of PVC localization in FIB

PVC means brightness differences in FIB (or SEM) images of structures having different electrical properties. The die isn't connected to any power supply or signal sources (passive) and is in most cases partially deprocessed. Failure localization with PVC is based on the fact that floating structures are charging up under the influence of the primary beam in FIB and SEM. In FIB this charging is always positive and greatly depends on the primary beam current. In the SEM the charging can be positive or negative depending on the primary beam energy and also depends on the primary beam current. This will be detailed below. Figure 1 shows the case of positive charging which is observed more often. Insulated structures are charging up positively under the influence of leaving secondary electrons. Immediately after that the majority of produced secondary electrons are prevented from leaving the sample by the electric field. These structures appear dark in the image because the electrons can't reach the secondary electron detector. Grounded structures do not charge and appear bright because of the high secondary electron yield.

If a structure that usually supposed to be floating appears bright, then it is in urgent suspicion to be faulty. On the other hand, if a structure that supposed to be grounded appears dark, this fact is also a strong indication for a fault. The first case is not just limited to structures that are floating a priori because of the design. Structures can be made floating by deprocessing as it is shown for a gate and its contact in Fig. 2.

Another possibility to make structures floating is cutting conducting lines with a FIB. If such a line is bright instead of dark, the unwanted substrate short can be found by

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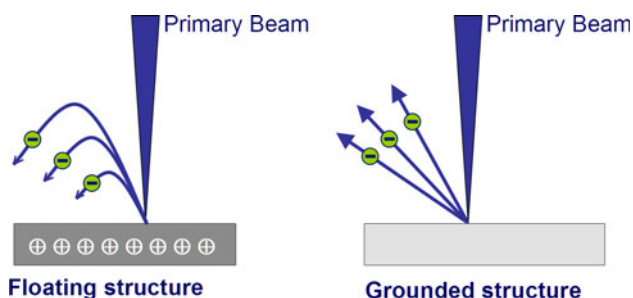


Fig. 1 Positive charging of insulated structures

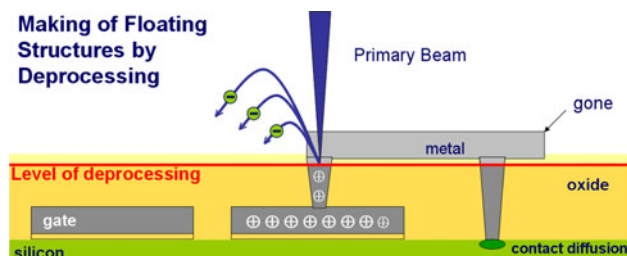


Fig. 2 Structures can be made floating by deprocessing

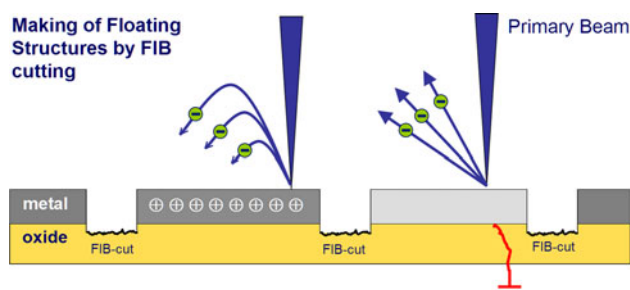


Fig. 3 Structures can be made floating by FIB cutting and faulty parts can be found

cutting the line into pieces as it is schematically shown in Fig. 3. The remaining bright part is bearing the fault.

For other failure localization issues it can be helpful to investigate grounded structures. Figure 4 shows that structures can a priori be grounded because of the design such as n diffusions in n-wells or p-diffusions in p-wells.

If one would investigate a structure that isn't connected to ground it can be grounded by milling a FIB hole through it followed by a metal deposition. The left structure in Fig. 5 is supposed to be dark because it is floating. If it isn't, there must be a short (thin line in the oxide) to the adjacent one which was purposely grounded for the localization.

The basic principles of passive voltage contrast generation can be summarized as follows: Floating structures can be made by deprocessing or by FIB cutting. They appear dark. Structures can be grounded from the outset by design or can be connected to ground by FIB. They appear bright.

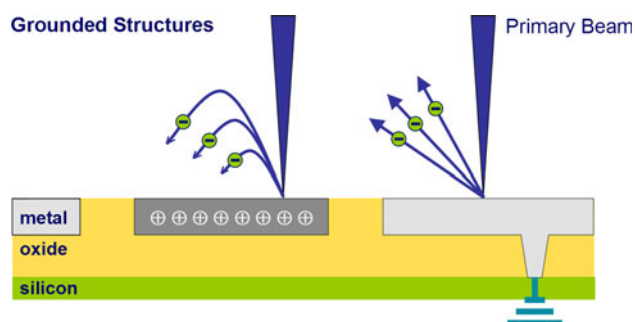


Fig. 4 A partially deprocessed sample with a structure (right) connected to ground by design

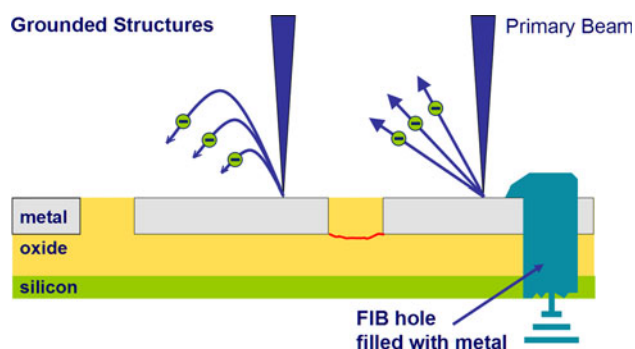


Fig. 5 A short between two adjacent floating structures can be detected by grounding one of them by FIB

2.2 PVC application examples and case studies

In Fig. 6 one part of a contact chain appears dark because it is floating due to a contact open. The bright part is grounded by design. In this context the term “grounded structure” means, that it has any contact to any substrate region or to a large network in contradiction to the conventional meaning where “grounded” indicates a resistance to mass potential of almost zero.

Figure 7 shows the application of a binary search strategy for localization of a line short in a large comb-serpentine test structure.

The serpentine had been grounded by drilling a FIB hole down to the substrate and filling it with Platinum by gas assisted deposition. The comb was isolated and was supposed to be dark but it wasn't because of a short to the serpentine. The comb was cut into two pieces, one piece turned dark, the other one stayed bright. Next the bright piece was cut into another two parts of which one remained bright (see top left). At the end of this procedure just one single horizontal comb finger was still bright. Now this finger was cut into two equal parts. The bright part was cut again and so on. Finally a little piece was isolated (see right micrograph) which was connected to the serpentine by a conducting particle.



Fig. 6 Open in a contact chain

Figure 8 shows a test array of several hundred deep trench capacitors. The electrical test showed that there was a leaking path but couldn't determine which capacitor was leaking. A simple FIB scan with medium beam current revealed clearly that there were three leaking capacitors at the upper edge of the array. They were the only ones which didn't charge up and thus didn't appear dark.

2.3 Advanced mechanisms of PVC generation

If diodes, capacitors and transistors are involved, PVC generation and interpretation is much more complicated. In the case of substrate contacts the doping has to be taken into account. When large structures are investigated, their capacitance plays an important role. Even transistors can affect the PVC generation when they are broken. In Fig. 9 the PVC generation at Source/Drain contacts (S/D) is shown. Gemmill et al. already described the VC at the 2 different basic types of substrate contacts, p^+ and n^+ diffusion in [8]. The diodes, formed by n^+ diffusion in p-wells

are reverse biased. The charging generated by the leaving secondary electrons can't flow down to the substrate. These contacts appear dark. The p^+ diffusions in n-wells are forming forward biased diodes. Here the positive charges can flow easily to ground, such contacts are bright.

But contacts connecting the wells (p-diffusion in p-well and n-diffusion in n-well) have also to be considered. They always appear bright because there is no diode. The darkest contacts are those connecting the gates because they are perfectly isolated from ground by the gate oxide.

In Fig. 10 the different contrasts of different contact types are to be seen.

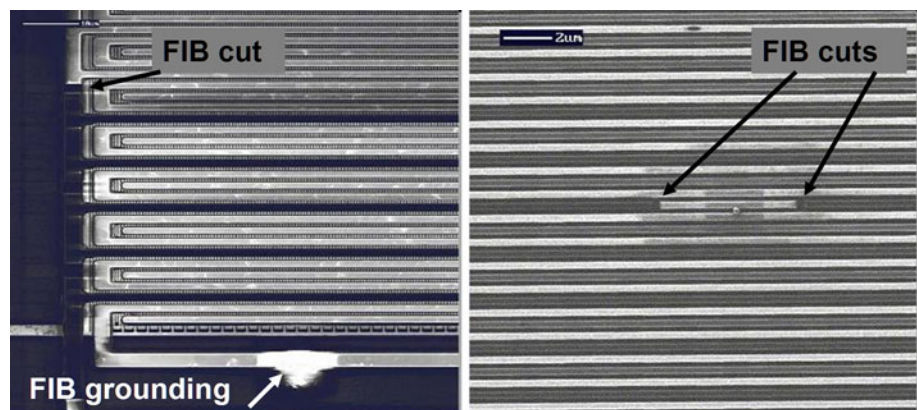
Figure 11 shows how the capacitance of a structure can influence the generation of PVC. If a structure is very large, e.g. a very long conducting line, then there are not enough charge carriers produced in order to raise the potential of this structure significantly. It stays bright like a grounded structure or becomes gradually dark because it is charging slowly during the image scan.

Figure 12 shows an equivalent circuit diagram for a structure having a certain capacitance C and a certain resistance R to ground. The current I_s is the sum of positive ions hitting the structure and of the secondary electrons leaving it. The solution of the corresponding differential equation delivers the voltage U_c which is building up at the structure.

The graphical representation shows the voltage curve in case of an endless charging. In reality, the charging only occurs during the time the ion beam scans over the structure, marked as t_{scan} . The charging is incomplete after this time but it is the larger the smaller the capacitance of the structure is. With reference to VC it means that a structure with a small capacitance often appears brighter than one with a larger capacitance.

The formula for the voltage U_c is also well suited for studying the influence of the leakage current from a structure to ground. These leakage currents can occur due to surface conductivity after deprocessing or due to the circuitry this structure is connected to.

Fig. 7 Binary search strategy for localization of a line short in a large comb-serpentine test structure



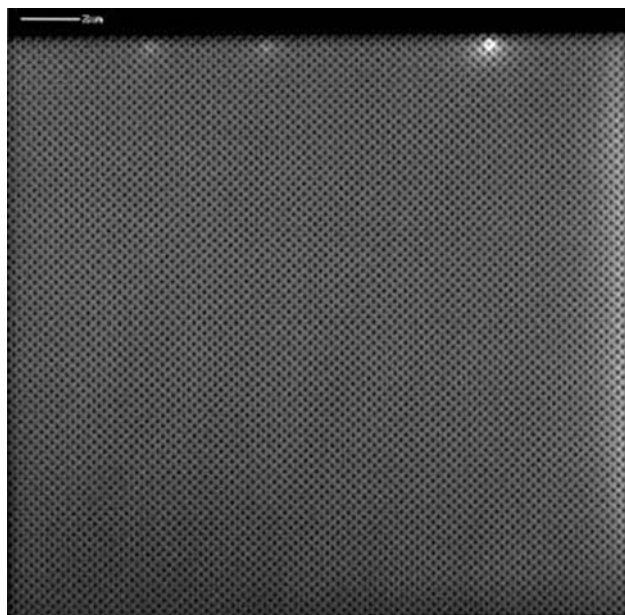


Fig. 8 Test array of deep trench capacitors in top down view with three leaking trenches

S/D contacts

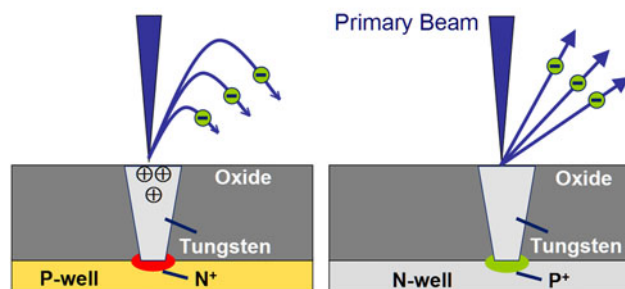


Fig. 9 PVC generation at S/D contacts

In Fig. 13 the graphical representation of $U_c(t)$ is to be seen for a fixed capacitance C but different values of the leakage resistance R . For low values of R the Voltage U_c

Fig. 10 Different contrasts of different contact types

VC in contact level of WL decoder of a DRAM product

Dark, gray and bright contacts can be explained by design interpretation.

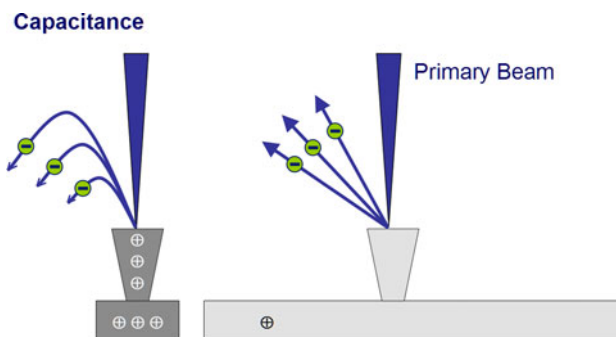
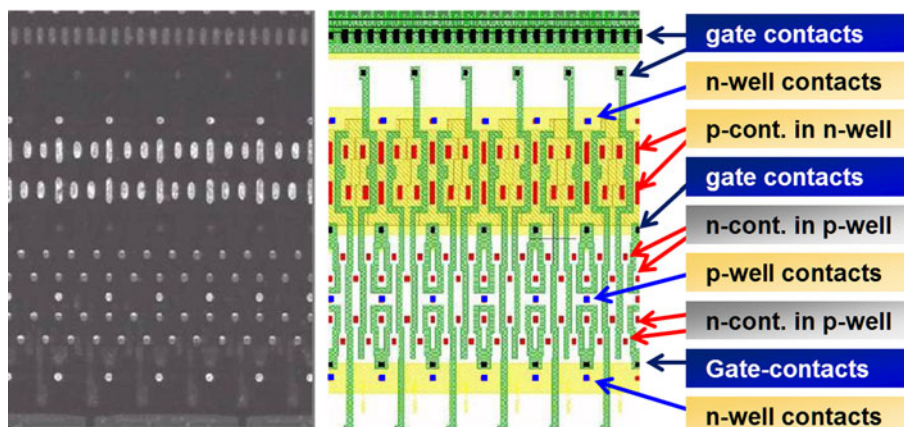


Fig. 11 Large structures (*right*) are like big capacitors which can't be charged sufficiently during the short time the ion beam scans over them

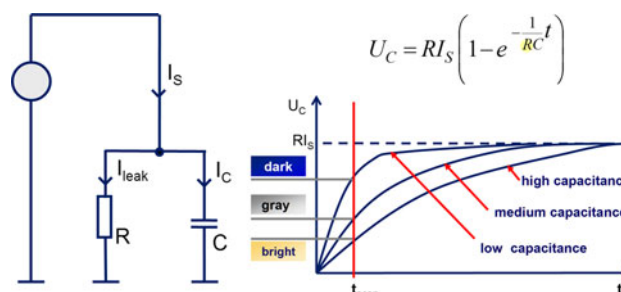


Fig. 12 Equivalent circuit diagram, solution of the corresponding differential equation and graphical representation for high, medium and low structure capacitances

can't build up to an amount which is necessary to prevent the secondary electrons from leaving. They can reach the detector resulting in a bright appearing structure. It was experimentally observed that R_{medium} is approximately 1 GOhm. Below this order of magnitude no dark structures can be observed.

At transistors another interesting PVC phenomenon could be seen. The contrast of s/d contacts changed depending whether the corresponding gate was grounded or not.

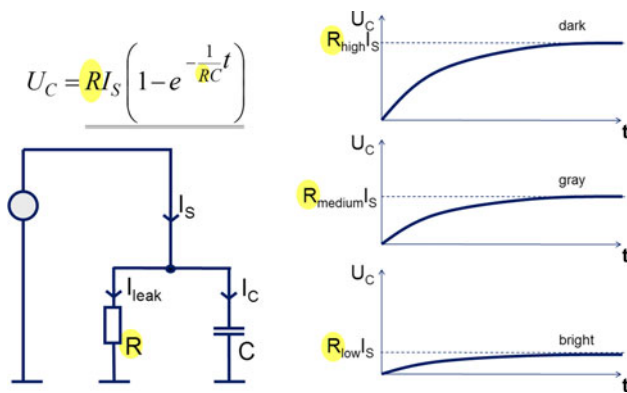


Fig. 13 Influence of leakage resistance to the voltage building up at the structure during ion beam scan

The left source contact in the transistor of Fig. 14 is normally expected to be dark because of the reverse biased diode under it. In reality it is bright because the charges can escape through the open transistor to the drain which is grounded. The gate was charged positively through the oxide during the ion beam scan and thus opened. When this gate conductor is grounded by a FIB cut through the gate oxide nearby (no metal fill is necessary) the source contact suddenly turns dark.

Figure 15 gives a summary of the advanced mechanisms of PVC generation.

2.4 Applications of advanced PVC methods and case studies

In Fig. 16 is demonstrated, that the capacity of a structure has to be taken into account.

The circles are marking contacts which are connecting gate conductor lines through landing pads of metal 0 and further contacts called CG. The sample is partially deprocessed. The contact in the second circle from top can't be seen because it is charged and shows a strong voltage contrast. On the other hand, the contacts in the other circles are also connected to floating structures (gate lines are

Implants	p-well	n-well
p-diffusion	bright	bright
n-diffusion	gray	bright
n-diffusion next to an open gate	bright	bright

Capacitance	high	medium	low
VC:	bright	gray	dark

Leakage Current	high	medium	low
VC:	bright	gray	dark

Fig. 15 Advanced mechanisms of PVC generation

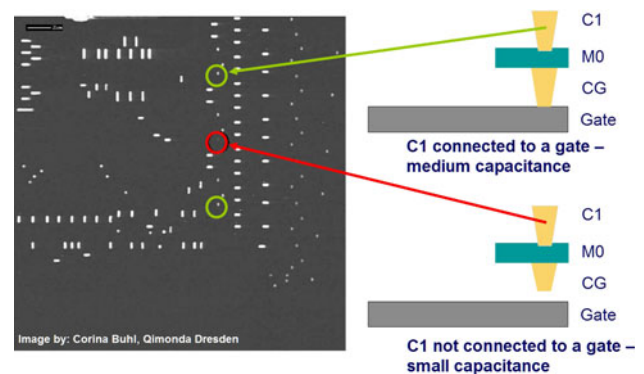
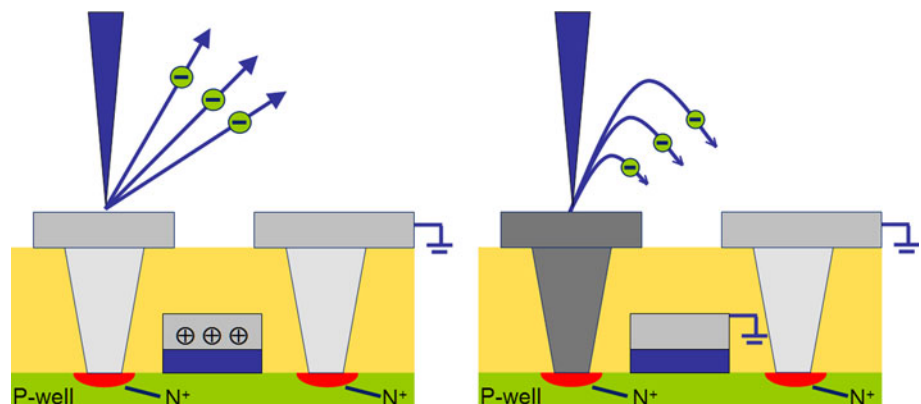


Fig. 16 Gate contact opens on a CMOS circuitry

always floating) and supposed to be dark but they aren't. The gate structures they are connected to are relatively large and have a considerable capacitance. The duration the contacts are exposed to the ion beam is too short to charge the whole structure. By this fortunate circumstance the open contact under the second circle from top could be found which could be confirmed by a subsequent cross section preparation in FIB as shown in Fig. 17.

The role of capacitance also can be studied in Fig. 18. One in four lines is floating and is relatively long whereas the others are grounded. The floating lines have such a high capacitance that they are charging gradually which can be seen during the scan which was running from top to

Fig. 14 The s/d contacts of n-MOS transistors are showing different PVC depending on whether the gate is opened or closed



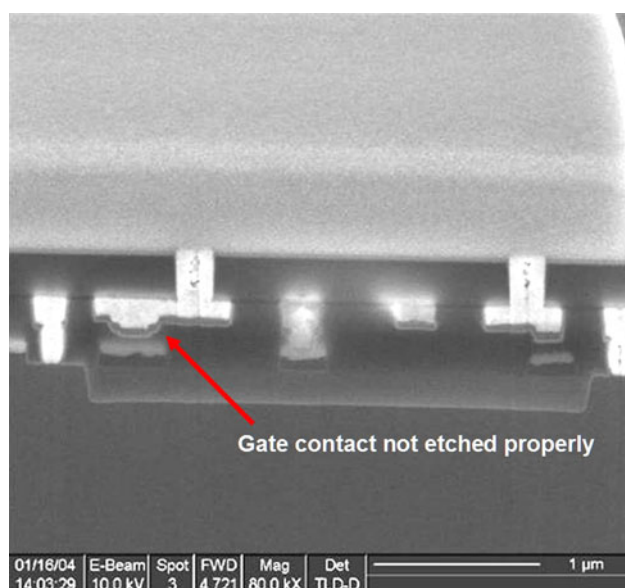


Fig. 17 Open gate contact localized by voltage contrast

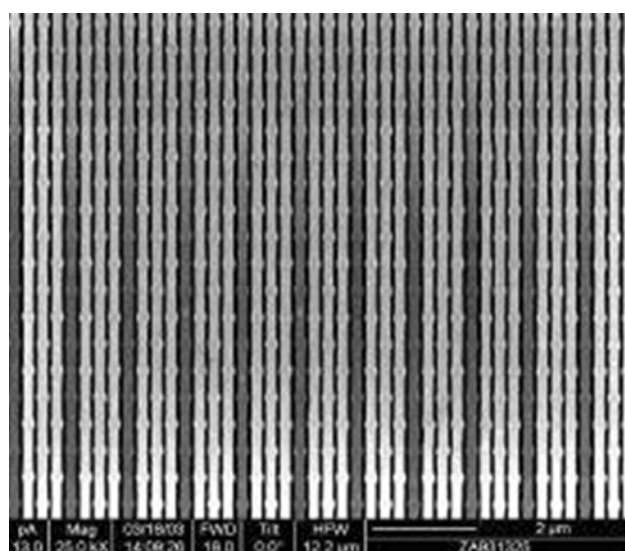


Fig. 18 Gradually charging of bit lines (one in four) having a smaller capacitance than the rest

bottom. The grounded lines are homogeneously bright because of their small capacitance.

The following failure localization case study derives benefit from the fact that npn transistors can be opened by charging up the gates positively and can be closed by grounding them.

Figure 19 shows a cross section through cell transistors of a deep trench DRAM. An ion beam scan over a partially deprocessed sample charges the gates through the remaining oxide and opens the transistors. The charges induced at the source contacts now can flow through the

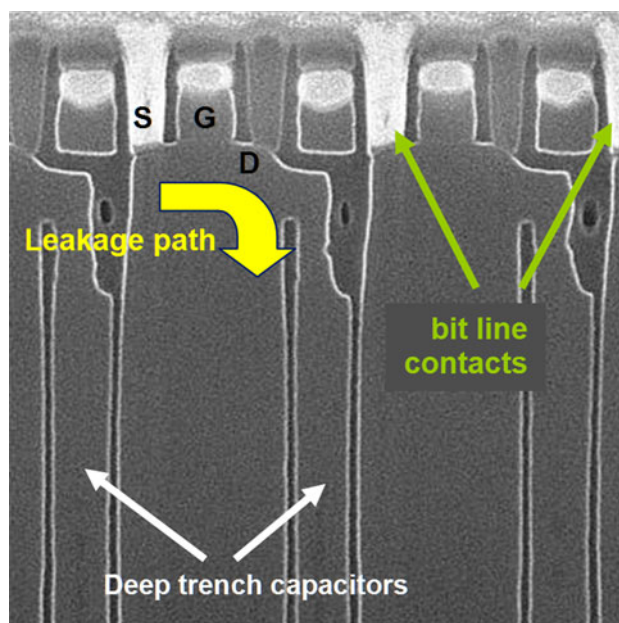


Fig. 19 Cross section through cell transistors of a deep trench DRAM product (source, gate and drain marked as S, G and D)

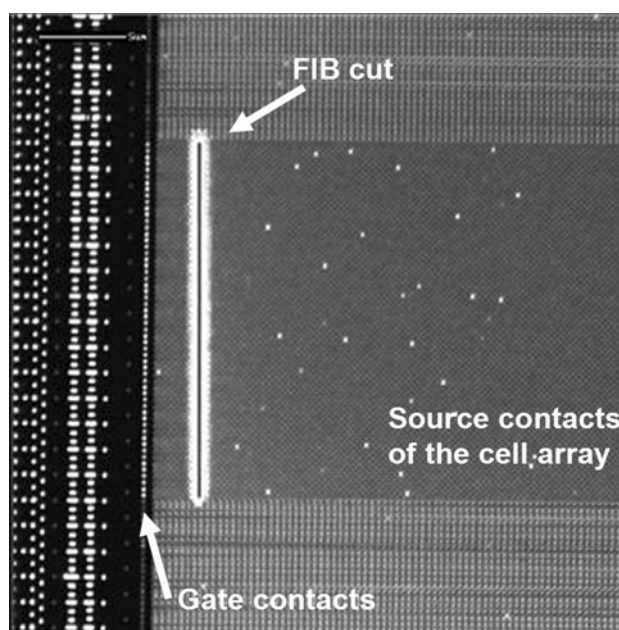


Fig. 20 Grounding of the buried horizontal running gate conductor lines with a FIB cut on the surface of a partially deprocessed DRAM cell array

channel down to the substrate in front of the capacitor as marked with the wide arrow.

In Fig. 20 a FIB line was milled through some of the gates connecting them with the substrate and thus grounding them. The marked gate contacts at the edge of the array are indicating this grounding clearly because they

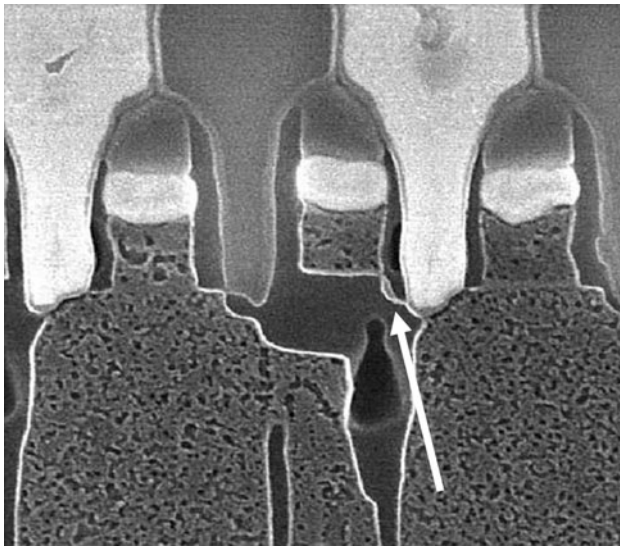


Fig. 21 The right source contact is shorted to a passing gate line by an etch residues

appear bright in contradiction to the gate contacts of that gate lines which hadn't been hit by the FIB cut.

The source contacts in the cell array turned dark. They are able now to charge up because of the closed cell transistors. However, some of them stayed very bright. A cross section through them as shown in Fig. 21 showed the reason for this. The right source contact is shorted by an etch residue to the passing gate line in between the two transistors. This gate line is FIB grounded like all gate lines in this area and consequently the gate contact is grounded too and appears bright.

What all these application examples clearly show is the fact that for an effective use of VC a detailed knowledge of CAD is required. Moreover, with CAD data the proper VC image of a faultless layer of a die can be predicted as it was shown impressively by Fredrickson in [9]. If the actual VC doesn't coincide with that prediction, the structure must be faulty.

2.5 Some experiences to share

Sometimes it is hard to achieve noticeable PVC in the Focused Ion Beam. Knowing the mechanisms behind PVC generation, one can derive some basic rules to get or to improve the contrast between charging and floating structures. The goal to be achieved is either to reduce the leakage currents or to improve the charge carrier generation.

1. The very first thing one should try is to increase the primary beam current, because it improves the net charging of the structure. It should be noted, however,

that there is always a tradeoff between damaging the structure by milling it away and the achievement of good PVC. In case of doubt the influence of a strong ion beam should be tested at an uncritical structure of the same kind. Figure 22 shows the influence of the primary beam current to the VC of a contact chain with several high resistivity contacts.

2. The next thing to try is to change the sample orientation. If structures are to be charged that are longer than wider, it is more effective to align their long side horizontally so that the horizontally scanning ion beam is hitting a big portion of the structure abruptly which can overcome possible leakage currents and can increase the chance to charge it up sufficiently. The easiest way to do it is to change the scan direction by software buttons instead of rotating the sample stage.
3. The scan speed also plays an important role. Slower scan speeds improve the chance of sufficient charging.
4. A tilted surface is emitting more secondary electrons than a horizontal one. This improves the charge carrier generation. Therefore stage tilting may help but it must be considered that FIB erosion will also increase.
5. When lines are to be FIB cut, one should always use the iodine gas injection because it prevents the redeposition of conducting material on the surface and thus it reduces the leakage currents (see Fig. 23).
6. Typical beam currents for the most localization issues are 10–50 pA. For low magnifications a current up to 100 pA can be used, for special problems at high magnifications 1–10 pA may be best.

3 PVC in SEM versus FIB

In Fig. 24 is explained why the charge generation in FIB is much stronger than in SEM.

In SEM, the incoming primary electrons and the leaving secondary electrons are partially compensating each other resulting in less charging than in FIB. Because this depends on primary beam current, sample material, sample orientation, magnification and other factors the charging in SEM can be both positive and negative, which can be an advantage in comparison to the FIB.

The accelerating voltage of the electron beam has the largest influence on the type of charging as it is shown in Fig. 25.

The threshold voltage E_1 could be determined experimentally and is approximately 0.5 kV and E_2 is around 2 kV. But it is only a rough approximation. Lai Li-Lung et al. have shown in [10] that these values strongly depend on surface topography and the material of the sample. In

Fig. 22 Influence of primary FIB beam current on VC

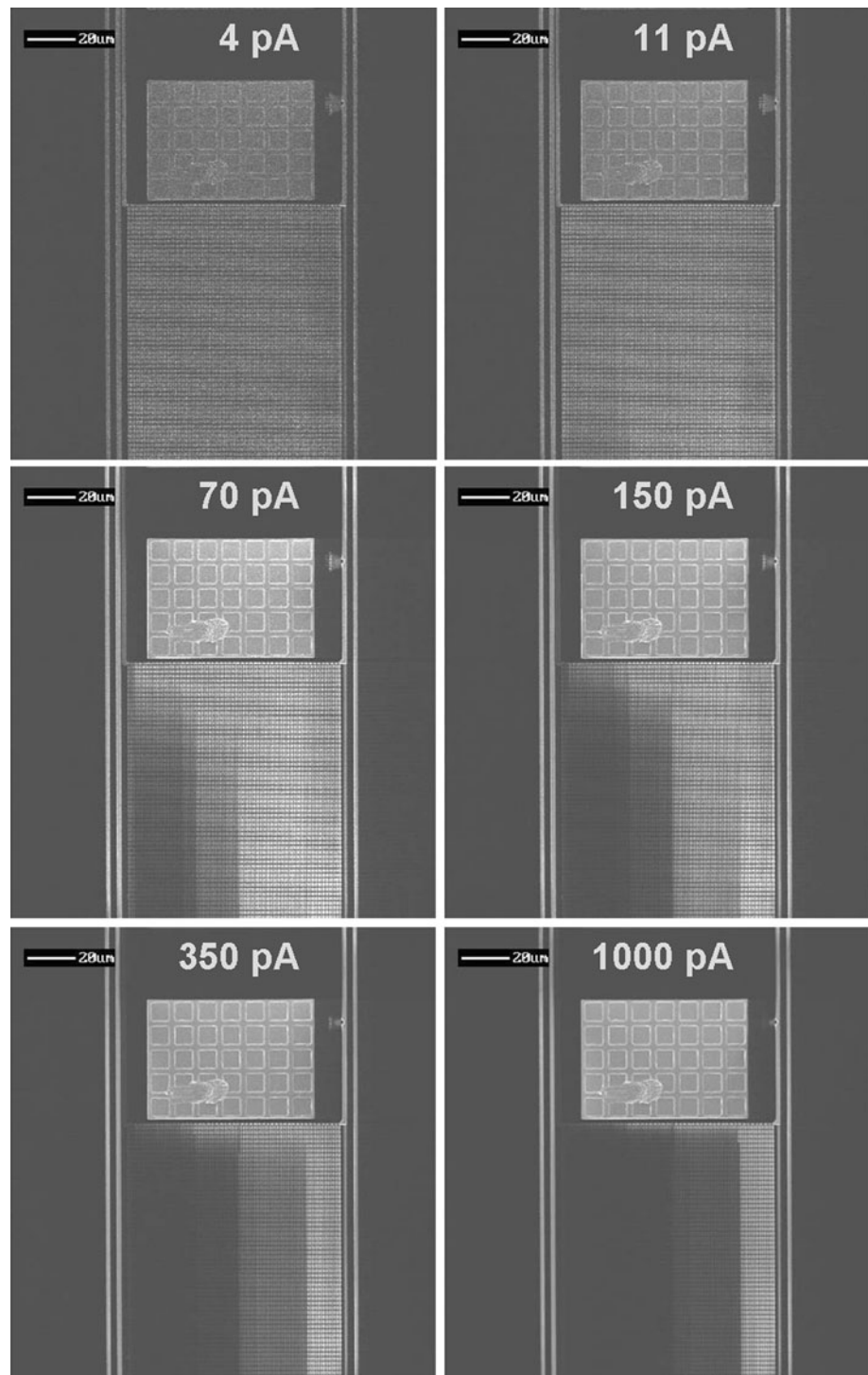


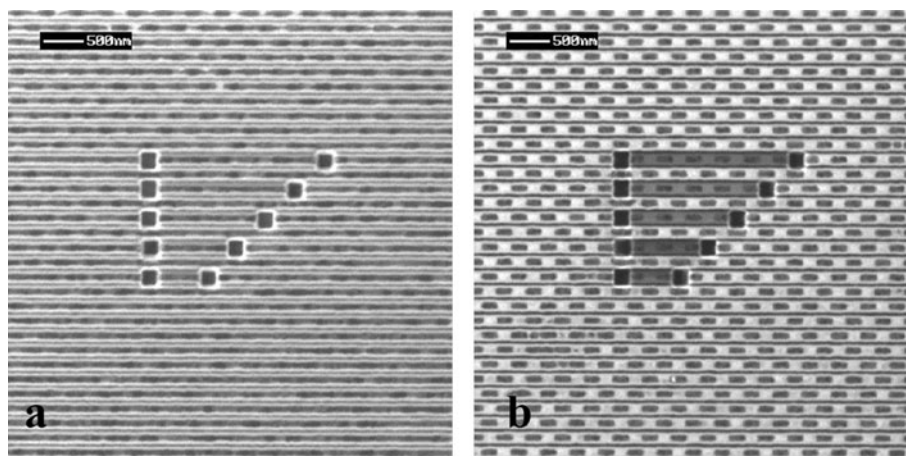
Fig. 26 SEM images with two different beam energies show different PVCs.

Depending on the problem to be solved either FIB PVC or SEM PVC is the method of choice. In Table 1 advantages and disadvantages of both methods are set against each other.

4 Inline e-beam inspection

A very important application of Passive Voltage Contrast in SEM is the inline e-beam inspection as described in [11–14] because it is the first and only FA technique consistently being used in the fab.

Fig. 23 Isolated pieces of conducting lines are scanned without (a) and with (b) iodine valve opened, the latter prevents leakage paths on the surface by resputtering and allows the isolated pieces to charge up positively



	FIB	SEM
Incident beam	positive charging by Ga^+ ions	negative charging by primary electrons
Leaving secondary e^-	positive charging	positive charging
Sum		(a) (b) (c)

Fig. 24 Comparison of Charging in FIB and SEM, in SEM both negative and positive charging is attainable depending on primary beam energy. In FIB just positive charging is possible but this is much stronger

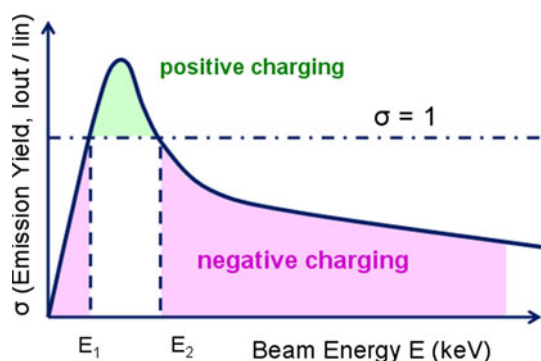


Fig. 25 Influence of the beam energy to the emission yield in SEM. The threshold voltages E_1 and E_2 depend on sample, sample material and SEM configuration like beam current, chuck to ground resistance, vacuum, etc

In an inline SEM a retarding field at the chuck provides low landing energies of electrons which are necessary for VC localization. The primary energy of the electrons right after having left the cathode is not affected by the retarding field and is as high as it is necessary for a high SEM

resolution. With a charge control plate mounted above the wafer both positive and negative surface charging can be detected (Fig. 27).

The example in Fig. 28 shows the detection of positive and negative charging of the same sample controlled by the charging plate. Only with the positive charging the contact open can be localized.

The example in Fig. 29 shows the opposite case. One of these unfilled contact holes wasn't deep enough and ended in the oxide. This just could be detected with the negative charging (The cross section on the right has been taken from this area after filling the holes with poly silicon).

The inline e-beam inspection is a powerful technique for accelerating the learning cycles in a wafer fab. Problems can be detected very early and yield predictions can be made before any electrical testing. However this technique is very time consuming. Although the process is highly automatized, the mapping of the contact test chains of a 300 mm DRAM wafer of the 70 nm node with 700 1-GBit dies for example takes approximately 6 h. It means that all wafers in line never can be inspected and a sampling will be always necessary.

5 Active voltage contrast (AVC)

With a nanoprobe in SEM/FIB as shown in Fig. 30 definite voltages can be applied to definite structures. This method became known as Active Voltage Contrast because the structures are set "active" when external voltages are applied to them. Applications of this principle were described already in 1999 by Campbell and Soden [15]. It is important to point out here that this principle was even earlier applied in the highly sophisticated method of Electron Beam Probing. Thong gave a good overview of it in [16]. Electron Beam Probing was once a very successful

Fig. 26 PVC in SEM, just at 0.4 kV noticeable PVC at the failing structure is to be seen, at 0.8 there is no difference between failing and reference structure

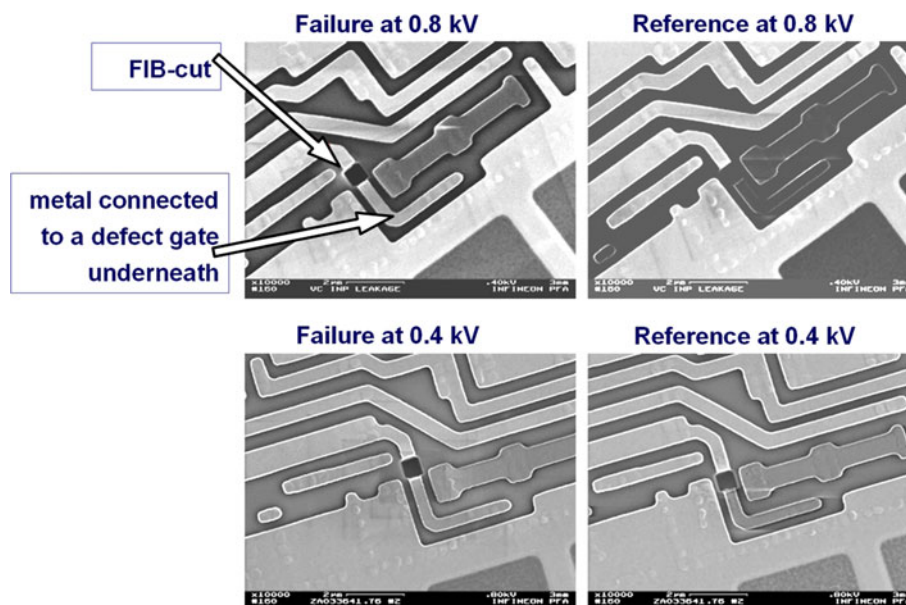


Table 1 Advantages and disadvantages of FIB and SEM PVC

	FIB	SEM
Visibility of PVC	Good	Poor
Charging	Just positive	Both
Cutting	Very helpful	Not possible
Sample degradation	Noticeable	Negligible
Marking of localized fails	Yes	No

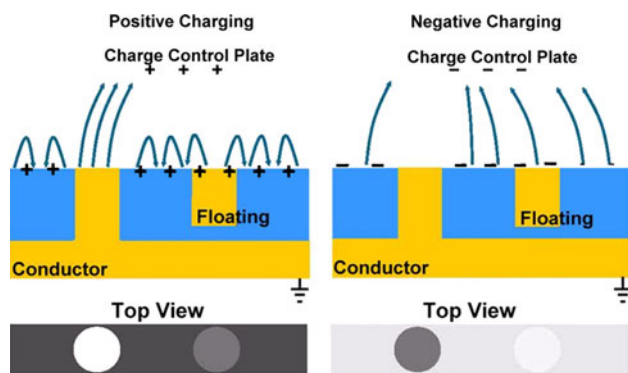


Fig. 27 Both positive and negative charges can be detected in inline e-beam inspection tools

technique for IC debug and Failure Analysis where AVC phenomena can be observed at the operating speed at a fully functional die. A decapsulated but not deprocessed IC or a single die on a fully processed wafer must be put into operation in a SEM. This means all I/Os and power supplies must be connected with a tester outside the chamber. The beam is directed to the interesting node. The changing surface potential of that node modulates the secondary

electron yield and the detector can register the waveform. With this dynamic active voltage contrast failure analysis can be performed at a very high level but an experienced FA engineer with a good knowledge of CAD design and IC functions is always necessary.

Unfortunately, a modern IC has a very high operating speed, is quite complex and the interesting nodes are not accessible due to the large number of metallization layers. These are the reasons why this method lost its significance during the past decade.

The AVC as described by the author in [17] and below has by far not the potential of the Electron Beam Probing but a considerably simpler instrumentation and can be performed by most SEM operators.

Sometimes contact chains are not completely open but have a high resistance far beyond the specification because of a partially open contact. The leaking current through this faulty contact prevents charging and its localization with PVC. This is shown in Fig. 31.

With an external voltage applied to such a chain as shown in Fig. 32 there are sufficient charges in the left part of the chain and at the faulty contact is a leap in potential according to Ohm's law.

An example for such a chain is shown in Fig. 33.

Another interesting example for AVC is shown in Fig. 34. To a single bit line of a memory array a positive voltage has been applied. Suddenly, the whole line turns black. Only the small part of it at the right end stays bright because it is isolated from the bias by an open.

Currently there are two basic types of in-chamber nanoprobe systems available—stage mounted ones and chamber mounted ones. Both have advantages and

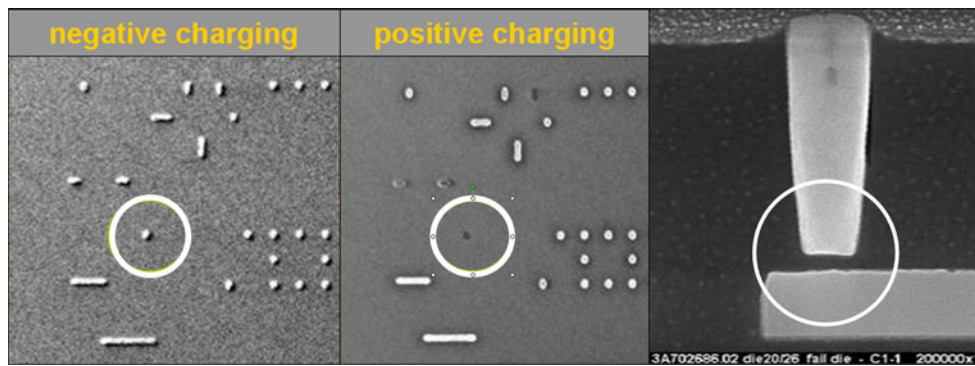


Fig. 28 Inline e-beam inspection. This contact open can be seen in positive charging only

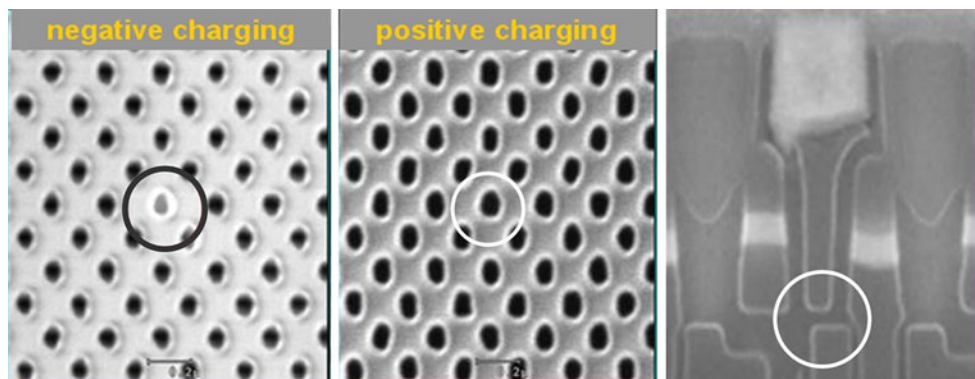


Fig. 29 Inline e-beam inspection. Contact holes that are etched not deep enough can be seen in negative charging only

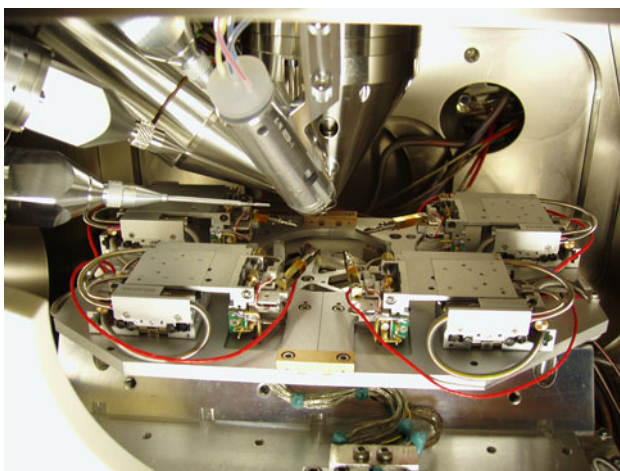


Fig. 30 4-Probe nanoprober, mounted on a stage of a Dualbeam FIB

disadvantages depending on the purpose they are to be used for. Chamber mounted system do not allow any stage movement after the probes have been lowered. It doesn't matter for device characterization purposes where 4, 6 or even 8 probes have to contact structures in a very small region.

For AVC sometimes test structures are contacted at their probe pads because it is easier than to do so at tiny

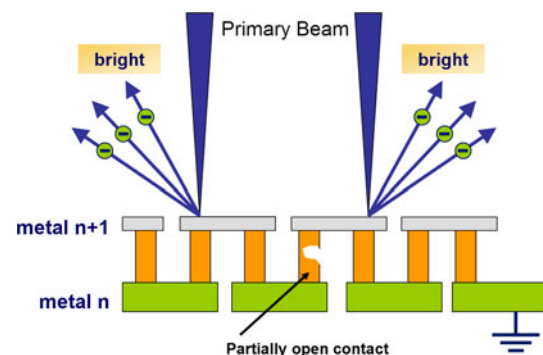


Fig. 31 The contact chain part left from the faulty contact can't be charged because charges can escape through it faster as they were generated by the primary beam, PVC localization just works on totally open contacts

conducting lines. Probe pads often are several hundreds of micrometers away from the structure itself. Here a stage mounted system is very helpful, because the pad is placed under the beam first in order to lower the tips under visual control and after that the stage is moved together with the tips until the test structure is under the beam in order to search for AVC.

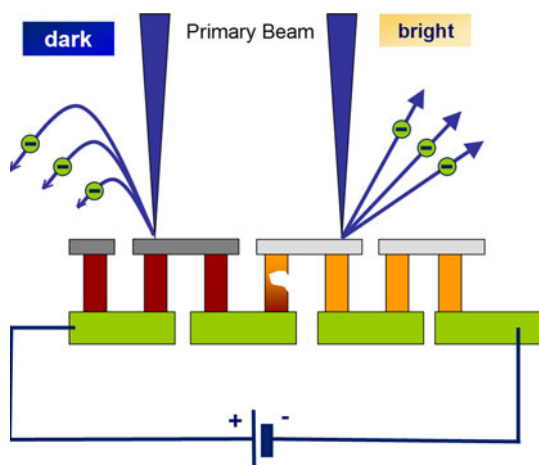


Fig. 32 Principle of Active Voltage Contrast. With an external voltage applied to a high-resistive contact chain different potentials emerge on the two parts of the chain resulting in a clearly distinguishable Voltage Contrast

Stage mounted systems have some restrictions regarding stage moving range and sample size. Some prober models do not allow the use of the load lock due to geometrical restrictions. On the other hand, this prober type can easily be used at any other tools of the same type, just an electrical feed through flange is necessary at each tool. In Table 2 advantages and disadvantages of both prober types are listed.

The number of probe tips for AVC application is not critical. It can be performed even with a single tip, two are sufficient whereas a device characterization requires 4, 6 or 8 tips.

Another interesting question is whether one should use a SEM or FIB for AVC localization issues. A SEM with 2 kV primary beam voltage can mostly be used. As an exception, when lines are to be cut, a FIB is needed. A tool with both SEM and FIB column would be ideal.

The last question to be discussed here is which voltages should be applied to the structures. Since most of the secondary electrons have energies of 2–4 eV and below, a

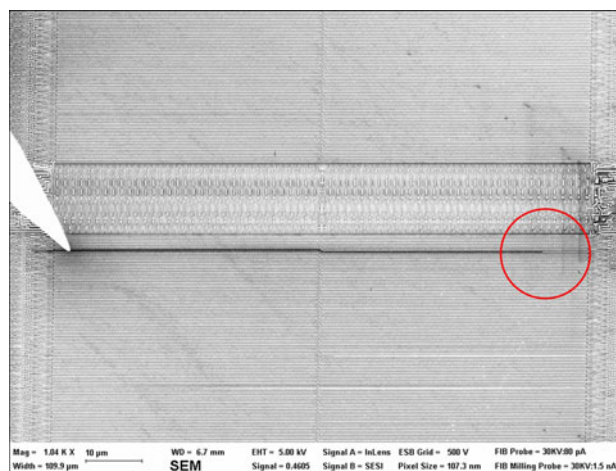


Fig. 34 Detecting an open in a memory bit line by applying a positive voltage to it

probing voltage of +5 V is o.k. A general rule for the probing voltage is “the higher the better” but at voltages above 7 V in a 60 nm CMOS process with 2 nm thermal gate oxide thickness an irreversible gate oxide damage could be observed preventing any further localization. In case of the new high-k gate oxides like TiN, TiAlN, Al_2O_3 or Hafnium Oxide this effect could be totally different and has to be studied in future.

6 Summary

For failure localization Voltage Contrast both in SEM and in FIB can be used. Both methods have their advantages and disadvantages.

Knowledge of all facts influencing the VC generation (capacitance, leakage, doping and circuitry) is very helpful for successful failure localization.

Active Voltage Contrast is more laborious but offers completely new opportunities for failure localization.

Fig. 33 With AVC one high resistance contact in a chain can be detected with both positive and negative voltage applied to the structure

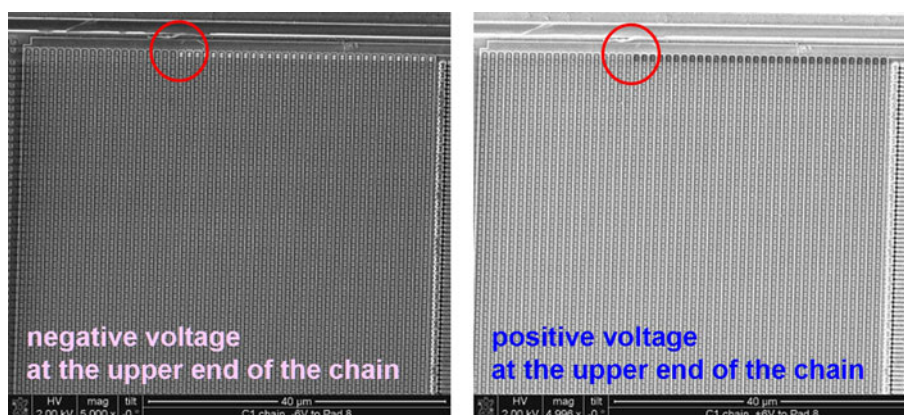


Table 2 Advantages and disadvantages of the two basic nanoprobe types

	Stage mounted	Chamber mounted
Stage movement	Stage movement with contacted probe tips	No stage movement with contacted probe tips
AVC	Any distance between probing area and failing area possible	Probing and failing area must be in the range of beam shift
	In large failing areas it can be searched for AVC by stage moving	No tilting
	Tilting in Dualbeam tools possible	
Device characterization	Well suited	Well suited
Restrictions	Stage moving range and sample size can be restricted	Nearly no influence to stage functionality
Flexibility	Easy to switch to another tool of the same kind (just feedthroughs must be on both tools)	Less flexibility
Time	For sample exchange the use of loadlock is not always possible. Chamber venting and evacuating takes time	No chamber venting unless no probe tip change is necessary

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